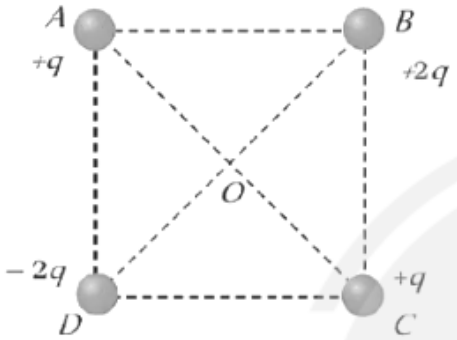
Lakshya JEE 2.0 (2025)

Physics

DPP: 2

Electric Charges and Fields

Q1 Four charges are arranged at the corners of a square ABCD, as shown in the adjoining figure. The force on the positive charge Q kept at the centre O is



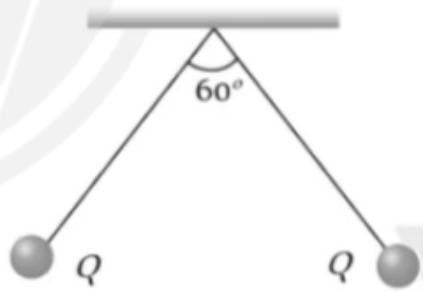
(A) Zero
(B) Along the diagonal AC
(C) Along the diagonal BD
(D) Perpendicular to side AB

Q2 A charge q is placed at the centre of the line joining two equal charges Q. The system of the three charges will be in equilibrium, if q is equal to
(A) $-\frac{Q}{2}$
(B) $-\frac{Q}{4}$
(C) $+\frac{Q}{4}$
(D) $+\frac{Q}{2}$

Q3 Dimensions of ε_0 are
(A) $\mathrm{M}^{-1}\ \mathrm{L}^{-3}\ \mathrm{T}^4\ \mathrm{A}^2$
(B) $\mathrm{M}^0\ \mathrm{L}^{-3}\ \mathrm{T}^3\ \mathrm{A}^3$
(C) $\mathrm{M}^{-1}\ \mathrm{L}^{-3}\ \mathrm{T}^3\ \mathrm{A}$
(D) $\mathrm{M}^{-1}\ \mathrm{L}^{-3}\ \mathrm{T}\ \mathrm{A}^2$

Q4 Consider two point charges of equal magnitude and opposite sign separated by a certain distance. The neutral point due to them
(A) Does not exist
(B) Will be in mid way between them
(C) Lies on the perpendicular bisector of the line joining the two
(D) Will be closer to the negative charge

Q5 Two small spherical balls each carrying a charge Q $= 10\mu\mathrm{C}(10$ micro-coulomb $)$ are suspended by two insulating threads of equal lengths $1\ \mathrm{m}$ each, from a point fixed in the ceiling. It is found that in equilibrium threads are separated by an angle 60° between them, as shown in the figure. What is the tension in the threads.



(Given: $\frac{1}{(4\pi\varepsilon)} = 9 \times 10^9\ \mathrm{Nm}^2/\mathrm{C}^2$)
(A) $18\ \mathrm{N}$
(B) $1.8\ \mathrm{N}$
(C) $0.18\ \mathrm{N}$
(D) None of the above

Q6 Two small conducting sphere of equal radius have charges $+10\mu C$ and $-20\mu C$

respectively and placed at a distance R from each other experience force F_1. If they are brought in contact and separated to the same distance, they experience force F_2. The ratio of F_1 to F_2 is

(A) $1:2$ (B) $-8:1$

(C) $1:8$ (D) $-2:1$

Q7 Two-point charges $+9e$ and $+e$ are at $16\ \mathrm{cm}$ away from each other. Where should another charge q be placed between them so that the system remains in equilibrium

(A) $36\ \mathrm{cm}$ from $+9e$

(B) $12\ \mathrm{cm}$ from $+9e$

(C) $24\ \mathrm{cm}$ from $+e$

(D) $12\ \mathrm{cm}$ from $+e$

Q8 The distance between charges $5\times10^{-11}C$ and $-2.7\times\ 10^{-11}C$ is $0.2\ \mathrm{m}$. The distance at which a third charge should be placed in order that it will not experience any force along the line joining the two charges is

(A) $0.44\ \mathrm{m}$ from $-\ 2.7\times10^{-11}\ C$ charge

(B) $0.65\ \mathrm{m}$ from $-\ 2.7\times10^{-11}\ C$ charge

(C) $0.556\ \mathrm{m}$ from $-\ 2.7\times10^{-11}\ C$ charge

(D) $0.350\ \mathrm{m}$ from $-\ 2.7\times10^{-11}\ C$ charge

Q9 An infinite number of point charges, each carrying $1\mu\mathrm{C}$ charge, are placed along the y-axis at $y=1\ \mathrm{m},\ 2\ \mathrm{m}, 4\ \mathrm{m}, 8\ \mathrm{m}\ldots\ldots\ldots\ldots$ The total force on a $1\mathrm{C}$ point charge, placed at the origin, is $x\times10^3\ \mathrm{N}$. The value of x, to the nearest integer, is _____.

[Take $\frac{1}{4\pi\varepsilon_0}=9\times10^9\ \mathrm{Nm}^2/\mathrm{C}^2$]

Q10 Two identical non-conducting solid spheres of same mass and charge are suspended in air from a common point by two non-conducting, massless strings of same length. At equilibrium, the angle between the strings is $\boldsymbol{\theta}$. The spheres are now immersed in a dielectric liquid of density $800\ \mathrm{kg\ m^{-3}}$ and dielectric constant 21. If the angle between the strings remains the same after the immersion, then

(A) Electric force between the spheres remains unchanged

(B) Electric force between the spheres reduces

(C) Mass density of the spheres is $840\ \mathrm{kg/m^3}$

(D) The tension in the strings holding the spheres remains unchanged

Answer Key

Q1	(C)	Q6	(B)
Q2	(B)	Q7	(B)
Q3	(A)	Q8	(C)
Q4	(A)	Q9	12
Q5	(B)	Q10	(B, C)

Android App | iOS App | PW Website